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Source-Scored Structural Initial Designs for Chance-Constrained Optimization of Ordered Policy Profiles

(Authors' names are not included for peer review)

Abstract. A policy with thousands of ordered settings is often a finely discretized rule, not a collection of unrelated decisions. We study how historical simulations can help choose the first few policies for a new chance-constrained problem of this kind. Our method starts from a public library of 64 simple profiles. Replicated source outcomes rank these profiles by objective and safety, while farthest-first selection chooses ten that also cover a low-dimensional cosine coordinate. The ten policies are fixed before any target outcome; any declared target optimizer and an independent verifier can then use them without changing the initial design. We prove cross-grid coordinate stability, finite-library coverage, recovery of separated source ranks, conditional coverage of an aligned target safe region, and exact familywise safety for the final shortlist. Across 480 paired task–resolution cells from 160 independently seeded latent tasks, source scoring achieves 91.7% feasible coverage and 47.3% certified deployment, versus 37.5% and 10.6% for outcome-free cosine maximin. The best source-free design wins in four adverse regimes and in a held-out electricity study. Historical simulation therefore helps only when profile geometry transfers and its cost is reused across targets.

Key words: simulation optimization, chance constraints, initial design, transfer learning, functional decisions, independent verification

1. Introduction

Many simulation decisions are rules before they are vectors. A reserve rule changes with forecast stress, an inventory rule changes over a planning horizon, and a resource rule changes across ordered operating states. A computer may store such a rule as hundreds or thousands of numbers, but those numbers are not unrelated knobs. They are samples of one ordered profile. Ignoring that fact can make the problem look much harder than the underlying decision really is.

Chance constraints make the first simulations especially important. If none of them reaches a feasible region, a surrogate has little evidence about where the safety boundary lies. Historical simulations from related systems can help, but only if the relevant policy shapes transfer. A shape that was safe in one system may be unsafe in another, and the cost of collecting and verifying historical evidence may exceed the cost of the new search itself. The useful question is therefore not simply whether history helps, but what part of history can be reused, under what assumption, and at what cost.

We focus on one concrete use of history: choosing the *initial design* for a new chance-constrained ordered-profile problem. We first construct a public library of 64 constant, monotone, smooth, piecewise, and oscillatory profiles without using outcomes. Replicated source simulations then rank these same profiles by objective and chance margin. The best compromise starts the design, and farthest-first selection adds profiles that cover a common cosine coordinate. The resulting ten policies are fixed and mapped to the target grid before the first target outcome. We call this a *source-scored structural initial design*.

The method keeps three jobs separate. The initial design decides where a new search begins. Any declared optimizer may add target evaluations. Finally, a short ordered list of candidates is fixed and tested with fresh simulations; if none passes, the procedure abstains. Figure 1 shows the change in viewpoint. We do not try to solve a large discretized space only by making the acquisition rule more elaborate. We first recognize the ordered profile, use history to choose among transparent candidate shapes, and reserve fresh target outcomes for deployment verification.

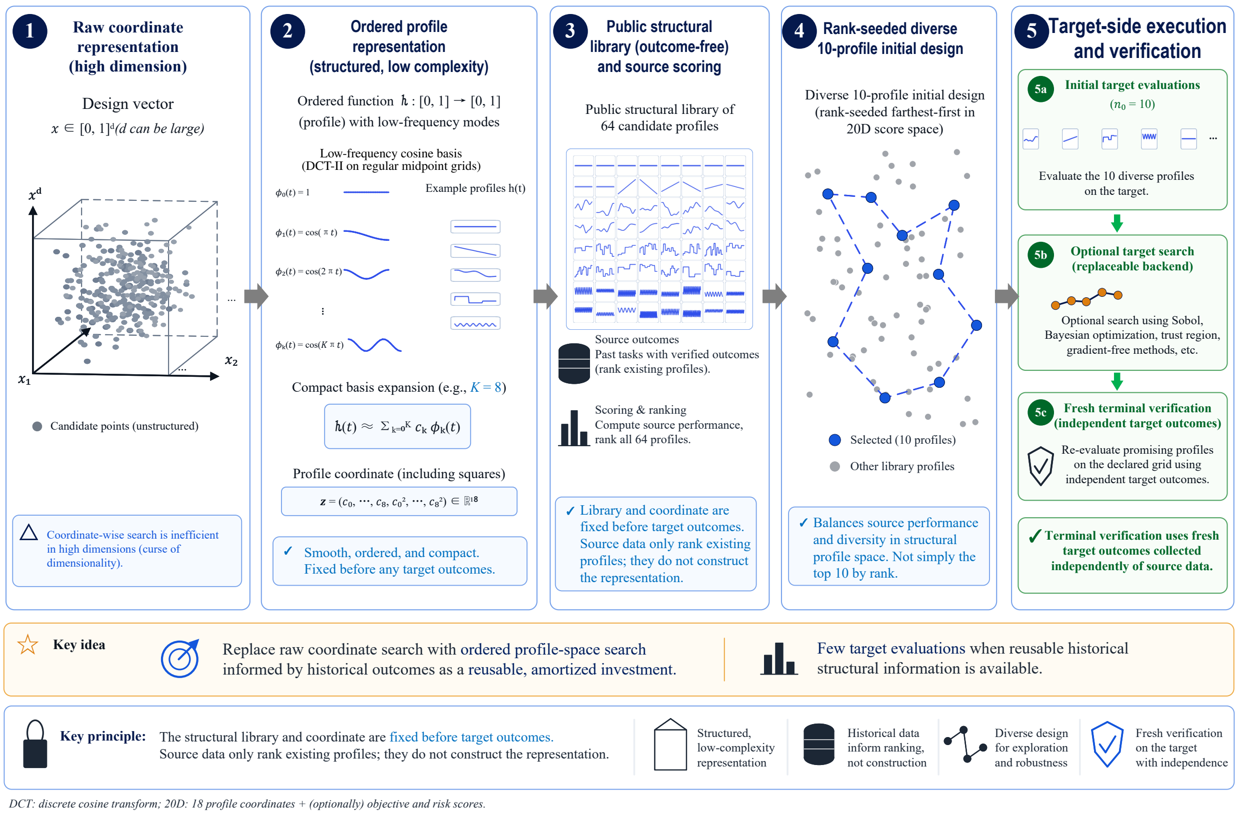

Figure 1 From high-dimensional coordinates to ordered profile-space search. The raw vector is read as samples of an ordered policy and summarized by low-frequency cosine coefficients. The public library and coordinate are fixed without outcomes. Source simulations rank existing profiles, while farthest-first selection prevents the ten-point design from collapsing onto a single source-favored shape. A replaceable optimizer may add target evaluations. Fresh target simulations then verify a fixed shortlist.

The paper makes three contributions.

1. We give a finite, reproducible way to turn source simulations into ten starting policies. The library, coordinate, weights, tie breaks, and target mapping are explicit, and the design uses no target outcome or hidden target geometry.

2. We explain when the small design can work. Grid refinement preserves the profile coordinate; farthest-first controls finite-library cover; finite source samples recover separated ranks; and structural cover reaches a target safe region when transfer error fits inside its safety margin. Fresh tests control false deployment. Lean 4 checks the finite implications, while source-target alignment remains an empirical assumption.

3. We test that assumption in eight randomized regimes, structured source-free comparisons, equal-cost experiments, and a region-held-out electricity study in which source scoring loses. Source, target-search, and verification calls are reported separately. The evidence therefore measures an

amortized design investment for related tasks, not a shortcut for one arbitrary high-dimensional problem.

Across 160 independently seeded tasks and three grid resolutions, source scoring improves feasible coverage and certification in aggregate but loses in several adverse regimes and in Energy. The 10,000-coordinate result shows stability under grid refinement, not a ten-evaluation solution to an arbitrary 10,000-variable black box.

2. Related Literature

Simulation optimization and noisy constraints. Simulation optimization studies expensive stochastic systems for which objectives or constraints are available only through simulation (Fu 2002, Amaran et al. 2016). Stochastic kriging explicitly separates response uncertainty from simulation noise (Ankenman et al. 2010), while constrained Bayesian optimization models noisy objective and feasibility observations jointly (Letham et al. 2019). Our question occurs one stage earlier: before fitting a target surrogate, which few ordered profiles should receive the first simulations?

High-dimensional and functional optimization. Trust-region methods and sparse axis-aligned priors can make Bayesian optimization more robust in large ambient spaces (Eriksson et al. 2019, Eriksson and Poloczek 2021, Eriksson and Jankowiak 2021). They do not by themselves encode that coordinates discretize one ordered function. Functional Bayesian optimization instead searches over function-valued decisions (Vien et al. 2018). Functional-data analysis likewise represents curves through common bases and treats discretization as observation of a function rather than unrelated coordinates (Ramsay and Silverman 2005). Our profile coordinate is less flexible but fully finite: it maps irregular or refined grids into a common cosine summary, then selects from a declared structural library. Nominal grid dimension and effective profile complexity are therefore reported separately.

Space-filling initial designs. Maximin and minimax distance criteria are classical model-free devices for covering an experimental region (Johnson et al. 1990), and space-filling designs are standard for deterministic computer experiments (Morris and Mitchell 1995). Gonzalez farthest-first gives a factor-two finite k-center guarantee (Gonzalez 1985). We use that algorithm on a finite profile library, but source ranks augment its selection metric. The generic-DCT control removes those ranks and is therefore the direct source-free space-filling comparison.

Transfer and historical-task priors. Few-shot deep kernels, hierarchical Gaussian-process priors, source ensembles, and meta-learned acquisitions transfer different objects across tasks (Wistuba and Grabocka 2021, Rothfuss et al. 2021, 2023, Wang et al. 2024, Tighineanu et al. 2022, Volpp et al. 2020, Pan et al. 2024). Related work also learns a search space from earlier hyperparameter tasks (Perrone et al. 2019) or transfers optimizer rankings (Feurer et al. 2018). Warm-start Bayesian optimization similarly treats earlier evaluations as reusable information (Poloczek et al. 2016). Our method transfers a smaller object: the source archive ranks a fixed public library, while the target surrogate and optimizer remain separate. This makes it possible to compare source scoring directly with the same structural library without outcomes and to count the cost of history explicitly. We also compare eight end-to-end transfer pipelines using the same source information and target budget.

Selection and independent verification. Chance-constrained ranking-and-selection procedures allocate simulation to identify a feasible best system while controlling selection error (Hong et al. 2015). Post-optimization ranking and selection can also "clean up" candidate solutions generated by a heuristic search (Boesel et al. 2003). We adopt this separation literally. Search produces an ordered shortlist; fresh Bernoulli feasibility replications test its members; and Bonferroni spending composes candidate-wise exact tests. The all-success rule is conservative, so we report power and verification cost in addition to false certificates. This differs from treating a surrogate's posterior feasibility probability as a deployment certificate.

Positioning. Without restrictions on the objective and feasible set, no finite target-label-free design can dominate all others, a point consistent with no-free-lunch results (Wolpert and Macready 1997). Our claim is therefore about experimental design for related ordered-profile tasks, not a distribution-free optimizer. Source-free structured designs, adverse regimes, an external negative control, and explicit cost accounting show where that claim does and does not hold.

3. Problem Setting and Available Information

3.1. Ordered policy profiles

Let $h : [0,1] \to [0,1]$ denote an ordered policy profile. A task q supplies ordered nodes

$$0 < t_{q,1} < \cdots < t_{q,d_q} < 1$$

and an integer implementation scale L_q. The deployed vector is

$$x_{q,i}(h) = \mathrm{round}\{L_q h(t_{q,i})\}, \qquad i = 1, \ldots, d_q.$$

Thus d_q is a grid resolution, not by itself the intrinsic dimension of the admissible profile class. Regular grids use $t_{q,i} = (i - 1/2)/d_q$; irregular grids are allowed when their ordering is declared. A schema may additionally declare channel order. No claim is made for an unordered vector with unknown coordinate semantics.

One simulator replication at x returns an objective observation $Y_q^f(x)$ and a scalar constraint observation $Y_q^g(x)$. We minimize

$$f_q(x) = \mathbb{E}[Y_q^f(x)]$$

subject to the chance constraint

$$p_q(x) := \mathbb{P}\{Y_q^g(x) \le \tau_q\} \ge 1 - \alpha_q. \tag{1}$$

The true feasible set is $\mathcal{F}_q = \{x : p_q(x) \ge 1 - \alpha_q\}$. In synthetic experiments, the experimenter reveals truth only after design, search, shortlisting, and verification are complete, and uses it only to score the finished decision.

3.2. Source and target information

Let $q_1, \ldots, q_m$ be source tasks and $q_\star$ a target task. Before the target opens, every source task evaluates the same finite normalized profile library $\mathcal{L} = \{h_1, \ldots, h_J\}$. For each pair (q, h), the source archive contains R ordinary replications of (Y_q^f, Y_q^g). The primary method uses $m = 2$, $J = 64$, and $R = 3$, hence

$$S = mJR = 384 \tag{2}$$

source simulator calls. The library is constructed without source or target outcomes; source data only determine which members are selected.

The source archive is supplied before the target problem opens. In the synthetic stress test its tasks come from the same declared regime as the target; in the external study they obey the stated region-holdout rule. The method ranks profiles within this archive. Choosing or purchasing related historical tasks from a larger repository is a separate upstream decision.

Before target search, the method may use the target's ordered grid, integer bounds, nominal dimension, and, in declared-schema experiments, channel order. It may not use a target outcome, feasibility label, noise parameter, optimum, safe-basin center, or verification sample. The primary source aggregation is domain blind: source tasks receive equal weight. Descriptor-conditioned weighting is evaluated only as a control.

3.3. Decision stages and costs

The initial-design step selects n_0 target policies. A target optimizer may then use a total of $N \geq n_0$ search calls and return an ordered list of at most K_s distinct candidates. This list and its order are fixed before fresh verification simulations begin. The final outcome is either a certified deployment or abstention.

We report the costs

$$C_{\mathrm{all}} = S + N + V, \qquad C_{\mathrm{amort}}(M) = \frac{S}{M} + N + V, \tag{3}$$

where V is the realized independent-verification cost and M is the number of target deployments sharing one source archive. "Equal preverification cost" means that $S + N$ is matched before V; it is not called equal total cost.

3.4. Endpoints

The primary search endpoint is whether the N target evaluations contain a truly feasible policy. The primary deployment endpoint is whether the verifier returns a truly feasible policy. We also report false certificates, verification calls, and a penalized loss that retains failures in the denominator. Conditional regret is reported only when a feasible policy exists and is not used alone to compare methods with different feasible counts. A separate fixed library defines synthetic regret and an unattainable finite-library upper bound; neither affects an implementable decision.

4. Source-Scored Structural Initial Design

The initial design has three steps: construct a public library of policy shapes, summarize each shape in a common coordinate, and use source outcomes to choose a promising but diverse subset. This section specifies each step.

4.1. Outcome-free profile library

All tasks share a deterministic library $\mathcal{L} = \{h_1, \ldots, h_{64}\}$ on 128 regular cell midpoints. Its seed is 20260808. It contains:

- ten constants with levels $0.05, 0.15, \ldots, 0.95$;
- six linear ramps with endpoint pairs $(.05, .95)$, $(.95, .05)$, $(.20, .80)$, $(.80, .20)$, $(.35, .65)$, and $(.65, .35)$;
- sixteen low-frequency cosine profiles, with rank drawn uniformly from $\{2, \ldots, 8\}$, coefficient $a_k \sim N(0, .12^2/k)$, and baseline drawn uniformly from $[.25, .75]$;

- sixteen piecewise-constant profiles with 3–12 blocks and levels drawn uniformly from $[.08, .92]$, followed by the declared deterministic smoothing;
- sixteen cosine profiles with frequency drawn uniformly from $\{9, \ldots, 40\}$, baseline in $[.30, .70]$, amplitude in $[.08, .25]$, and a uniform phase.

Values are clipped to $[0, 1]$. Duplicate profiles are skipped, and stable profile identifiers follow construction order. This mixture is a structural prior. It is public, generated before any outcome, and is not claimed to span all useful policies.

4.2. Grid-invariant profile coordinate

For ordered nodes $t_1 < \cdots < t_d$, let $0 = e_0 < e_1 < \cdots < e_d = 1$ be their Voronoi cell edges, where $e_i = (t_i + t_{i+1})/2$ internally. Write $\Delta_{ik} = \sin(\pi k e_i) - \sin(\pi k e_{i-1})$. For the piecewise-constant reconstruction of h, define

$$c_0(h) = \sum_{i=1}^{d} h(t_i)(e_i - e_{i-1}), \tag{4}$$

$$c_k(h) = \frac{\sqrt{2}}{\pi k} \sum_{i=1}^{d} h(t_i)\Delta_{ik}, \quad k = 1, \ldots, K. \tag{5}$$

These are exact integrals of the cosine basis over the reconstruction cells, not midpoint approximations to the basis. On regular midpoint grids they coincide with DCT-II coefficients up to normalization; on irregular grids we use their exact cell-integral cosine analogue. With $K = 8$ and $\kappa = .25$, the profile coordinate is

$$\phi(h) = \left(\tilde{c}_0, \ldots, \tilde{c}_K, \tilde{c}_0^2, \ldots, \tilde{c}_K^2\right), \qquad \tilde{c}_k = \frac{c_k}{1 + \kappa k}. \tag{6}$$

It has dimension 18 and contains no cross-products. Each coordinate is standardized over the 64-profile library before selection.

Here c_0 is the profile's average level, while $c_1, \ldots, c_8$ describe its broad upward, downward, and oscillatory shapes. Their squares add a simple measure of mode strength without introducing every pairwise interaction. The cell integrals are what let the same shape receive a comparable description on a 200-, 1000-, or 10,000-point grid.

4.3. Replicated source scores

For source task q, profile j, and replication r, let Y_{qjr}^f and Y_{qjr}^g be the two simulator outputs. Define

$$\widehat{f}_{qj} = R^{-1} \sum_{r=1}^{R} Y_{qjr}^f, \tag{7}$$

$$\widehat{m}_{qj} = \overline{Y}^g_{qj} + z_{1-\alpha_q} \max\{s^g_{qj}, 10^{-3}\} - \tau_q, \tag{8}$$

where s^g_{qj} is the sample standard deviation with divisor $R-1$. Within each task, deterministic average-tie percentile ranks map the 64 objective means and margins to $[0,1]$, with lower values preferred. Let r^f_{qj} and r^g_{qj} denote those ranks. The primary domain-blind scores are

$$r^f_j = \frac{1}{m}\sum_{q=1}^{m} r^f_{qj}, \qquad r^g_j = \frac{1}{m}\sum_{q=1}^{m} r^g_{qj}. \tag{9}$$

Whether $\widehat{m}_{qj} \le 0$ in every source task is logged as a diagnostic but does not filter the library. Descriptor-conditioned source weights are a secondary control and are not part of the primary method.

The margin combines the observed constraint mean and scale in the same units as the threshold: a negative value favors feasibility. Percentile ranks then remove differences in objective and constraint scales across source tasks.

4.4. Rank-seeded farthest-first selection

Let z_j be the standardized version of $\phi(h_j)$. This 18-dimensional coordinate contains structural profile information only. Selection uses the 20-dimensional augmented coordinate

$$\eta_j = (z_j, r^g_j, r^f_j) \tag{10}$$

with Euclidean distance. The first center is

$$j_1 \in \arg\min_j \{.5 r^g_j + .5 r^f_j\}, \tag{11}$$

breaking ties by r^g_j, then r^f_j, then stable profile identifier. For $\ell = 2, \dots, n_0$,

$$j_\ell \in \operatorname*{arg\,max}_{j \notin \{j_1,\dots,j_{\ell-1}\}} \min_{s<\ell} \|\eta_j - \eta_{j_s}\|_2, \tag{12}$$

with the lowest library index breaking an exact tie. The primary design uses $n_0 = 10$. Each selected 128-node profile is linearly interpolated onto the declared target nodes, multiplied by L_q, clipped, and deterministically rounded to an integer policy.

Selecting the ten best source ranks alone could return ten nearly identical profiles. Farthest-first avoids that collapse: source performance chooses the starting point and influences distance, while each later point must add new coverage.

Table 1 Frozen primary method specification.

Component	Specification
Decision class	Ordered bounded profile $h : [0,1] \to [0,1]$ on a declared grid
Library	64 outcome-free profiles; 128 reference nodes; seed 20260808
Source archive	2 tasks $\times$ 64 profiles $\times$ 3 replications = 384 calls
Profile coordinate	Frequencies 0:8, penalty .25, diagonal squares, 18 dimensions
Source score	Equal-task average of objective and Gaussian chance-margin percentile ranks
Selection	First center $.5r^g + .5r^f$; nine Gonzalez farthest-first centers
Metric weights	Profile coordinates standardized; safety and objective rank weights both 1
Target map	Linear interpolation to declared nodes; deterministic integer rounding
Target information	Grid, bounds, dimension, and declared ordering; no outcomes or oracle quantities

The two coordinates play different roles. The structural coordinate z says what a profile looks like. The augmented coordinate η also says how that profile performed on source tasks, so it determines which profiles are selected. The target coverage theorem uses only z: projecting η back to z cannot increase Euclidean distance. We therefore do not pretend that a source percentile rank has a target counterpart.

4.5. Target search and verification

The initial design does not depend on a particular target optimizer. The primary comparison sets $N = n_0 = 10$, so no sequential search can blur the effect of the ten starting policies. Functional SCBO, SAASBO, and native transfer methods are evaluated separately.

Given noisy target records, the neutral protocol freezes up to three distinct shortlist entries in this order: best observed-objective empirical feasible policy, observed safe-interior policy, and minimizer of

$$Y^f(x) + 2\,[Y^g(x) - \tau]_+, \tag{13}$$

followed by objective-ranked fallbacks if needed. Fresh simulator replications then test shortlist members sequentially. Candidate j with budget v_j is certified only if all v_j constraint replications satisfy the threshold and $(1-\alpha)^{v_j} \le \delta/K_s$. The primary budgets are $(80, 80, 80)$, $K_s = 3$, and $\delta = .05$. Failure to certify produces abstention, not a deployment.

4.6. Source-free comparison designs

The closest comparison is *generic DCT maximin*. It uses the same 64 profiles, the same ϕ, the coordinate-space medoid as its first center, and farthest-first for the remaining nine, but no outcomes. Random low-frequency, natural three-block, and raw Sobol designs separate other forms of structural prior. The finite-library upper bound is unattainable and is used only to interpret the completed experiments.

Thus source data choose among profiles that already exist; they do not learn the library itself. Whether this scoring improves on generic structure is a question for the experiments, not an assumption of the method.

5. Why the Design Can Work: Finite-Sample Guarantees

The method uses only ten target policies, so its structural assumption must do real work. The results below trace that work from representation to deployment. Proposition 1 first shows that a policy has nearly the same coordinate when the implementation grid is refined. Theorem 1 then shows that farthest-first gives a controlled cover of the finite library. Proposition 2 states which source rankings survive finite simulation noise. Theorem 2 turns structural cover into a feasible target policy when a source-supported profile lies close enough to a sufficiently deep target safe region. Finally, Theorem 3 controls false deployment using fresh target simulations.

This chain also makes the limitation visible. A fine grid does not hurt if it describes the same ordered profile, but no theorem can rescue a target whose safe region is absent from, or badly distorted relative to, the structural library. The results are finite and conditional; they are not a guarantee for arbitrary high-dimensional black boxes.

5.1. Profile-grid consistency

Let $c_k(h) = \int_0^1 h(t) b_k(t)\, dt$, where $b_0(t) = 1$ and $b_k(t) = \sqrt{2}\cos(\pi k t)$ for $k \geq 1$. Let $\widehat{h}_d$ be a reconstruction from d ordered samples.

Proposition 1 **(Coefficient stability)**. *If $\|h - \widehat{h}_d\|_\infty \leq \epsilon_d$ and $\|b_k\|_\infty \leq B_k$, then*

$$|c_k(h) - c_k(\widehat{h}_d)| \leq \epsilon_d B_k.$$

If h is L_h-Lipschitz and $\widehat{h}_d$ is the regular-midpoint piecewise-constant reconstruction, then $\epsilon_d \leq L_h/(2d)$. Consequently the error of the frequency-penalized coefficient is at most

$$\frac{L_h B_k}{2d(1+\kappa k)}.$$

Linear interpolation from a reference grid of size d_{ref} has the analogous bound $L_h/(2d_{\mathrm{ref}})$ before deterministic rounding.

The bound says that nominal dimension disappears once the grid is fine enough to reconstruct the same profile. It justifies comparisons across discretizations of one ordered rule; it does not turn an unordered vector into a function. The proof is the integral triangle inequality followed by the nearest-node or linear-interpolation error.

5.2. Why ten diverse profiles cover the library

For any coordinate u on the finite library, define the covering radius of $A \subset \mathcal{L}$ by

$$r(A;u) = \max_{h\in\mathcal{L}} \min_{a\in A} \|u(h) - u(a)\|_2.$$

THEOREM 1 **(Farthest-first finite k-center bound).** *Let A_k be the k centers produced by Gonzalez farthest-first from any declared first center, and let $r_k^\star$ be the smallest covering radius attainable by any k library members. Then*

$$r(A_k;u) \le 2r_k^\star.$$

The standard witness proof uses the k centers and one farthest unselected point. Two of these $k+1$ pairwise-separated witnesses must share an optimal cluster. The implementation emits this witness certificate for every design. For the implemented $\eta = (z, r^g, r^f)$, Euclidean projection gives

$$r(A_k;z) \le r(A_k;\eta) \le 2r_k^\star(\eta). \tag{14}$$

Thus source ranks can steer selection without erasing structural diversity. Equation (14) gives a checkable cover in profile space; it does not claim that source ranks have the same meaning on the target.

5.3. What finite source samples can rank

Let s_{qj} denote either a true source objective statistic or a true source chance-margin statistic and $\widehat{s}_{qj}$ its replicated estimate.

PROPOSITION 2 **(Separated-rank recovery).** *Suppose $|\widehat{s}_{qj} - s_{qj}| \le r_q$ simultaneously over a finite source library. Every pair satisfying $|s_{qj} - s_{q\ell}| > 2r_q$ has the correct estimated order.*

For the Gaussian simulator used in the registered synthetic study, let $R \ge 2$, $\nu = R-1$, and let $q_{\nu,p}$ be the p quantile of χ^2_ν. For one profile with true mean μ and scale σ, define

$$r_\mu(\delta_\mu) = \sigma\sqrt{\frac{2\log(2/\delta_\mu)}{R}}, \tag{15}$$

$$a_\nu(\delta) = 1 - \sqrt{q_{\nu,\delta/2}/\nu}, \tag{16}$$

$$b_\nu(\delta) = \sqrt{q_{\nu,1-\delta/2}/\nu} - 1, \tag{17}$$

$$r_\sigma(\delta_\sigma) = \sigma \max\{a_\nu(\delta_\sigma), b_\nu(\delta_\sigma)\}. \tag{18}$$

The normal mean tail and Cochran identity $(R-1)S^2/\sigma^2 \sim \chi^2_{R-1}$ imply, with probability at least $1-\delta_\mu-\delta_\sigma$,

$$|\widehat{m} - m| \le r_\mu(\delta_\mu) + z_{1-\alpha} r_\sigma(\delta_\sigma), \qquad \sigma \ge 10^{-3}. \tag{19}$$

The variance floor is harmless here because $a \mapsto \max\{a, 10^{-3}\}$ is one-Lipschitz. Allocating the two tail probabilities across the finite set of tasks, profiles, and score heads gives the simultaneous radius required by Proposition 2. This result deliberately leaves nearly tied profiles unresolved: three replications cannot reliably order them. The sensitivity experiment therefore varies replications and library size instead of treating the source order as known. The normal and chi-square sampling laws supply the probability bounds (Cochran 1934); Lean checks the finite variance identity, the effect of the variance floor, the union of bad events, and the final rank implication.

5.4. When library coverage reaches a target safe region

Let $m_q(h)$ be the target chance margin, with $m_q(h) \le 0$ implying feasibility. Let $z_q^\star$ be an ideal target structural coordinate and $\widehat{z}$ the implemented structural projection of η. The following statement exposes the assumption that source and target share useful profile geometry; it imposes no target interpretation on (r^g, r^f). Write $d_q(h,h') = \|z_q^\star(h) - z_q^\star(h')\|_2$.

Assumption 1 (**Aligned profile geometry**). *There exist a target safe-center profile $h_q^\star$, a library member h_s, and constants $\gamma_q, \Delta_q, \epsilon_\eta, L_q \ge 0$ such that*

$$m_q(h_q^\star) \le -\gamma_q,$$
$$d_q(h_s, h_q^\star) \le \Delta_q,$$
$$\|\widehat{z}(h) - z_q^\star(h)\|_2 \le \epsilon_\eta,$$
$$|m_q(h) - m_q(h')| \le L_q d_q(h,h').$$

The third inequality holds for every profile used below.

Theorem 2 (**Aligned atlas coverage**). *Let $\mathcal{A}_{n_0}$ cover $\mathcal{L}$ within radius $r^\eta_{n_0}$ in augmented η. By* (14) *it covers the same support within that radius in $\widehat{z}$. Under Assumption 1, if*

$$L_q\left(r^\eta_{n_0} + \Delta_q + 2\epsilon_\eta\right) \le \gamma_q, \tag{20}$$

then $\mathcal{A}_{n_0}$ contains a target-feasible policy.

The condition has a direct reading. The ten profiles must cover the library closely; the source-supported shape must not move too far on the target; the implemented coordinate must approximate the target-relevant geometry; and the safe region must be deep enough to absorb all three errors. Frequency shifts, irregular grids, and misspecification increase Δ_q or ϵ_η, while a shallow feasible region decreases γ_q. In those cases the theorem stops, as it should. The complete triangle-inequality proof appears in the Online Supplement.

Figure 2 gives the geometric interpretation. The symbol ϵ_η denotes error between the implemented projected structural coordinate $\widehat{z}$ and the ideal target coordinate $z_q^\star$; it does not transfer the source-rank components (r^g, r^f) to the target. The circle labeled r_{n_0} depicts the structural-space upper bound obtained by projecting the augmented-coordinate cover $r_{n_0}^\eta$.

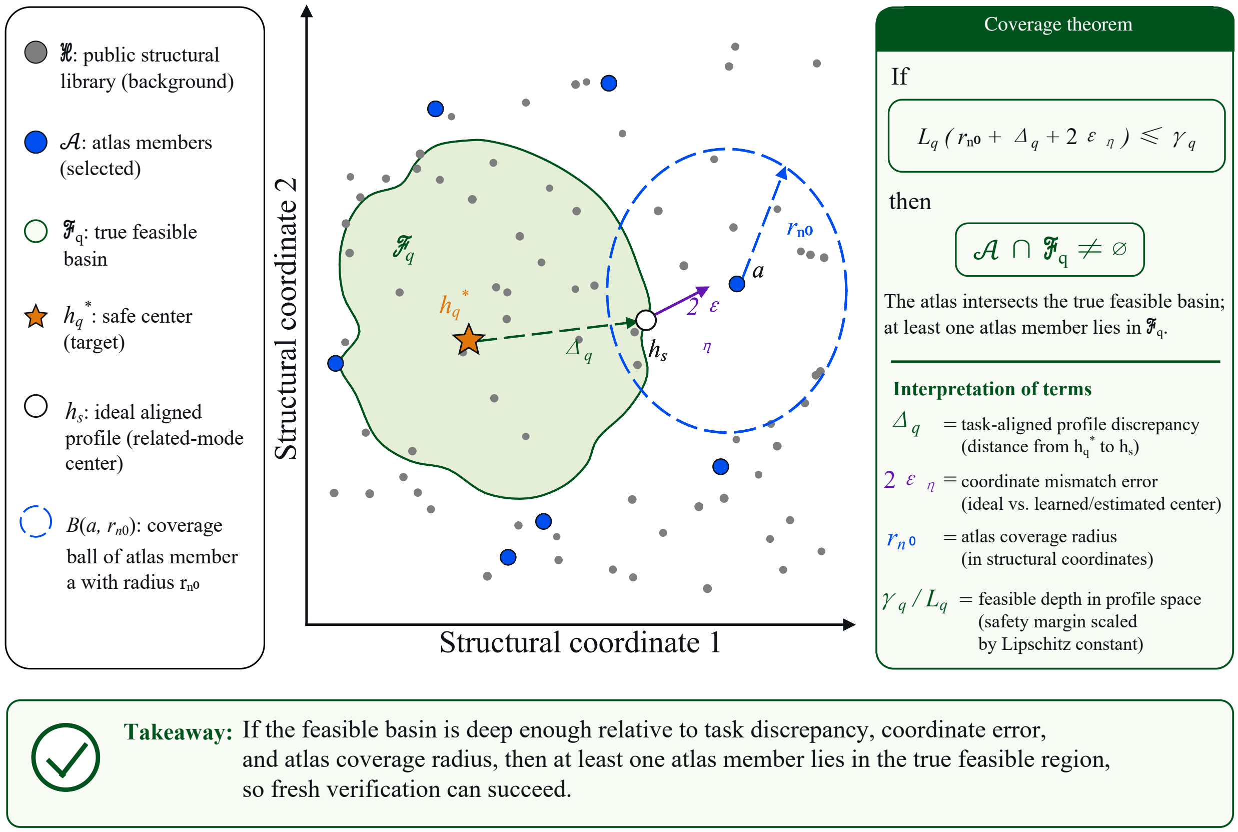

Figure 2 Schematic geometry of atlas coverage in structural profile space. Gray points represent the public profile library and blue points the selected atlas. The target has an unknown feasible basin $\mathcal{F}_q$ containing a safe center $h_q^\star$. A source-supported library profile h_s lies within task discrepancy Δ_q of the safe center; projected coordinate mismatch contributes at most $2\epsilon_\eta$; and an atlas member a covers h_s within the projected upper bound r_{n_0}. Hence $L_q(r_{n_0} + \Delta_q + 2\epsilon_\eta) \leq \gamma_q$ implies $\mathcal{A} \cap \mathcal{F}_q \neq \varnothing$. The diagram illustrates the coverage theorem and is not an empirical two-dimensional projection of a target instance.

The preceding theorem concerns one target. To estimate how often the selected design succeeds over the declared task population, we next treat a design hit as a bounded random outcome. The confirmatory generator uses eight fixed, equally weighted regimes, not one pooled IID law. At a fixed resolution d, let regime s have n_s independent held-out task seeds, weight w_s with $\sum_s w_s = 1$, and hit indicators $H_{si} \in [0,1]$. Define the task-law hit rate $p = \sum_s w_s \mathbb{E}[H_s]$, its stratified estimate $\widehat{p} = \sum_s (w_s/n_s) \sum_{i=1}^{n_s} H_{si}$, and $v_w = \sum_s w_s^2/n_s$. Weighted Hoeffding concentration, which does not require identical distributions across strata, gives with probability at least $1-\delta$,

$$p \geq \widehat{p} - \sqrt{\frac{v_w \log(2/\delta)}{2}}. \tag{21}$$

Here $w_s = 1/8$ and $n_s = 20$. The deterministic component `task_seed mod 5` is an induced latent category whose realized counts need not be balanced; independence and boundedness, rather than identical distributions, are sufficient. The same 160 latent task seeds are crossed with all three resolutions, so (21) is applied separately at each d; the pooled 480-cell rate is descriptive. The probability refers only to a prospective repetition of the registered stratified generator and requires the atlas to be frozen before held-out tasks. It is not a guarantee for arbitrary future domains.

5.5. Independent terminal safety

Fix a shortlist before verification. Candidate j receives v_j fresh iid Bernoulli indicators $Z_{jr} = 1\{Y^g_{jr} \le \tau\}$ and is certified only when $\sum_r Z_{jr} = v_j$ and

$$p_0^{v_j} \le \delta_j, \qquad p_0 = 1 - \alpha, \tag{22}$$

where $\delta_j = \delta / K_s$ in the primary protocol.

THEOREM 3 **(Candidate and familywise validity).** *For any fixed unsafe candidate with true feasibility probability $p_j < p_0$,*

$$\mathbb{P}\{\textit{candidate } j \textit{ is certified}\} = p_j^{v_j} \le p_0^{v_j} \le \delta_j.$$

For a frozen shortlist, regardless of dependence among the candidate tests,

$$\mathbb{P}\{\textit{an unsafe policy is deployed}\} \le \sum_{j=1}^{K_s} \delta_j \le \delta.$$

Sequentially stopping at the first certificate does not alter this union-bound argument because the shortlist and all budgets are fixed before testing.

For $p_0 = .95$, $K_s = 3$, and $\delta = .05$, the minimum all-success budget is 80 per candidate. Its power at true feasibility probability p is p^{80}; this is only 0.017 at $p = .95$. Validity therefore does not imply that the test will often certify, and Section 7 reports both validity and power. The exact Clopper–Pearson lower bound is used only for the preregistered sensitivity calculation (Clopper and Pearson 1934). A complete proof of the familywise statement appears in the Online Supplement.

5.6. Human-readable and machine-checked proofs

The Online Supplement gives ordinary mathematical proofs of the three central statistical and transfer results, followed by a map from manuscript claims to Lean files. Profile consistency, finite k-center, rank recovery, aligned coverage, task-law concentration, binomial verification, and budget implications are formalized in Lean 4 without `sorry`, `admit`, or a project-defined `axiom`. Machine checking confirms that the stated conclusions follow from the stated premises. It cannot establish that Assumption 1 holds in Energy or in an unspecified future domain; that remains an empirical question.

Table 2 Registered stress regimes. The target generator hides its basis, center, safe radius, and task seed from every algorithm.

Regime	Default rank	Principal departure
Aligned low frequency	8	Source and target share low-frequency support
Growing effective rank	24	More active coefficients at fixed target budget
Frequency support shift	8	Target frequencies shifted beyond source support
Coordinate permutation	8	Ordered values are permuted before implementation
Irregular grid	8	Nonuniform declared nodes
Piecewise smooth	12	Blockwise basis rather than cosine basis
Sparse high frequency	16	Few active high-frequency coordinates
Misspecified target	24	Target center independent of source centers

6. Experimental Design

We fixed the confirmatory protocol before observing its outcomes. A compact registry records the method version, experimental cells, failures, and analysis endpoints. Every implementable method sees target truth only after its deployment decision, when truth is used to compute evaluation metrics.

6.1. Randomized ordered-profile stress test

The primary study uses eight fixed, equally weighted regimes. Within each regime, independent deterministic seed streams generate held-out problem instances. Table 2 separates grid resolution from the complexity of the underlying policy shape.

For each regime, we draw 20 independently seeded latent target tasks and cross the same seeds with $d \in \{200, 1000, 10000\}$. This gives 160 independent latent task seeds and 480 correlated task–resolution cells per design. Each cell is paired across six designs: source-scored, generic DCT maximin, random low frequency, natural three-block, raw Sobol, and an explicitly labeled finite-library upper bound. The primary implementable matrix has $8 \times 3 \times 20 \times 5 = 2400$ cells; including the oracle gives 2880. Every design uses $n_0 = N = 10$, so the primary comparison is a pure initial-design intervention. The source-scored arm pays 384 source calls; source-free arms pay none. All use the same independent shortlist verifier.

For each synthetic target regime, the two supplied source tasks are generated from that same regime with the role set to source. This is a regime-matched transfer experiment, not a source-retrieval experiment. Algorithms receive neither target outcomes nor hidden target geometry, but the experimenter has already supplied a related archive. The Energy region holdout is the explicit archive-mismatch control.

The primary chance level is $\alpha = .05$, and task safe mass is calibrated to .08 on a separate fixed library that is not used for design. Regret is measured against the best feasible policy in that evaluation library. The $d = 10000$ cells refine the grid while holding the underlying problem fixed unless the regime explicitly changes rank.

6.2. Sensitivity and information controls

An 8640-cell one-factor-at-a-time matrix varies active rank, α, safe mass, n_0, number of source tasks, profiles per task, source replications, retained frequency, frequency penalty, source-rank metric weights, and the first-center safety weight. These experiments describe sensitivity; they are not a hyperparameter search performed after seeing the primary results. A separate 1280-cell matrix crosses declared versus schema-blind profile mapping with equal versus descriptor-conditioned source weights. The latter weighting is excluded from the primary method regardless of its result.

6.3. Cost and optimizer controls

The equal-preverification-cost matrix compares 384 source plus 10 target calls with 394 target-only calls for each source-free design. The same independent verifier is applied afterward and all-in realized calls include its cost.

Target-only functional SCBO searches an 8-coefficient cosine space with the published constrained trust-region posterior-sampling backend (Balandat et al. 2020, Eriksson and Poloczek 2021). We test coefficient counts 4, 8, and 16 at $N = 13$, and the primary 8-coefficient version at $N = 394$. One $N = 394$ task in the irregular-grid regime returned duplicate canonical posterior samples; it remains a failed, unsuccessful cell.

Eight native transfer pipelines (Safe F-PACOH, RGPE, stacked transfer GP, multi-task GP, FSBO, HyperBO, MetaBO, and MALIBO) receive the same 384-call source archive, $n_0 = 10$, $N = 20$, and terminal verifier, but retain their own prior, representation, initialization, and acquisition logic. This is an end-to-end transfer comparison, not a common-atlas backend comparison.

6.4. Electricity-reserve negative external control

The external study uses the Open Power System Data time-series release (Open Power System Data 2020). A policy maps 1000 ordered forecast-stress quantiles to reserve responses over a 168-hour operational horizon. Energy V3 includes 18 markets in five geographic regions and five algorithmic seeds per held-out market. All source markets from the target's region are excluded. The source arm uses 768 source calls, $N = 13$ target calls, and the same public profile library. Generic DCT, random low frequency, natural constant, raw Sobol, and target-only functional SCBO provide comparisons without source scoring.

Fresh verification windows are sampled from a held-out empirical calendar split. The exact-binomial claim is therefore conditional on the declared replication distribution, not a theorem about an unobserved future year. After the decisions, we also check chronological blocks and physically nonoverlapping windows. These checks are descriptive because markets share calendar data and only five regions exist. Energy V2 is retained as a separate historical specification; V3 is the final external control. No Energy result was used to modify the initial design.

6.5. Inference and failure accounting

For randomized profiles, a latent task seed within a fixed regime and resolution is the generalization unit; paired bootstrap intervals resample these seeds. Task-law concentration is applied separately at each resolution with regimes as fixed strata. Rates pooled over all three resolutions describe 480 task–resolution cells but are not treated as 480 independent tasks. Simulation seeds within a fixed external market measure algorithmic repeatability, while geographic regions are the conservative Energy comparison units. We do not pool seeds as if they were independent domains. Exact sign tests are Holm-adjusted within their registered families, but effect sizes and intervals receive priority over very small p-values. Every timeout, duplicate-sampling failure, and false certificate remains in its denominator.

Primary endpoints are feasible coverage, certified true-feasible deployment, false certification, and penalized loss. We additionally report source, search, verification, unamortized all-in, and $M = 20$ amortized calls; conditional regret is secondary.

7. Results

7.1. Aggregate randomized-task performance

Table 3 and Figure 3 report the primary randomized task law. The source-scored design contains a truly feasible policy in 91.7% of task–resolution cells and yields a certified true-feasible deployment in 47.3%. Generic DCT maximin reaches 37.5% and 10.6%; raw Sobol reaches 80.2% and 30.0%. Penalized loss is 0.163 for source scoring, 0.746 for generic DCT, and 0.337 for Sobol. Thus source outcomes add value beyond the shared structural library on this declared task mixture.

The paired source-minus-generic-DCT differences in certified success are 0.406 (bootstrap 95% CI [.325, .488]) at $d = 200$, 0.356 ([.275, .438]) at $d = 1000$, and 0.338 ([.256, .419]) at $d = 10000$. The similar differences across grids show that the result survives profile refinement under this task law. They do not show increasing difficulty with d: the $d = 10000$ experiment changes resolution, not the underlying policy class.

Table 3 Primary randomized profile results. Each row contains 480 paired task–resolution cells: 160 independently seeded latent tasks crossed with three grid resolutions.

Initial design	Tasks	Feasible (%)	Certified (%)	False cert.	Penalized loss
Generic DCT maximin	480	37.5	10.6	3	0.746
Natural blockwise	480	83.8	19.4	0	0.278
Random low frequency	480	50.8	13.5	0	0.610
Raw Sobol	480	80.2	30.0	2	0.337
Source-scored atlas	480	91.7	47.3	0	0.163

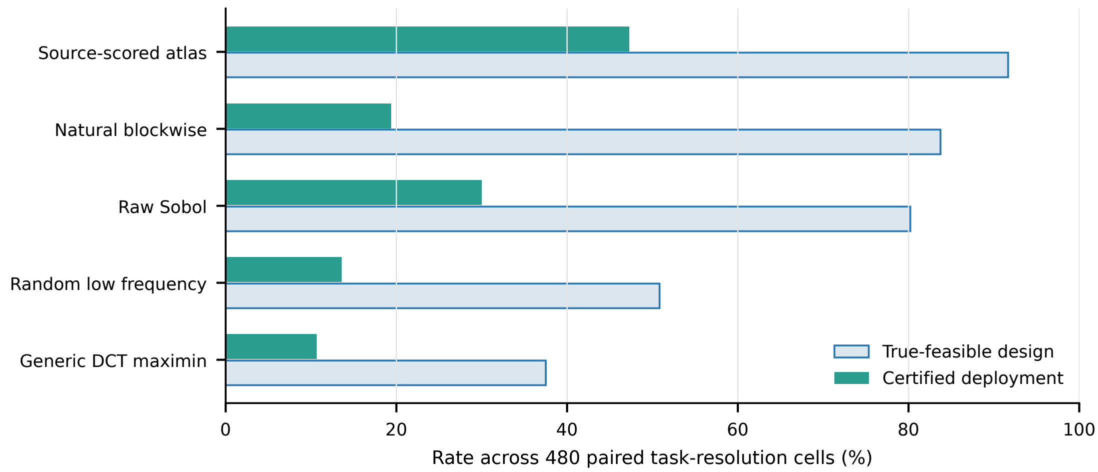

Figure 3 Aggregate feasible coverage and independently certified deployment. Bars aggregate task–resolution cells; inferential comparisons are paired within resolution and do not treat the three grids as independent tasks.

7.2. Where source scoring fails

The aggregate advantage is not uniform. Table 4 and Figure 4 preserve every adverse regime. Raw Sobol outperforms source scoring in aligned low frequency and irregular grids; natural structure is slightly better under sparse high-frequency activity; and neither method certifies useful policies under full misspecification. Source scoring is strongest for permutation, piecewise smoothness, growing rank, and the registered frequency shift. In the sparse-high-frequency regime its feasible coverage falls from 20/20 tasks at $d = 200$ to 0/20 at $d = 10000$, despite unchanged target budget. This is direct evidence against a generic high-dimensional claim.

The source arm records no false certificates in 480 task–resolution cells. Generic DCT and raw Sobol record three and two, respectively. Target-only functional SCBO at $N = 13$ certifies 15 of 480 primary cells with three false certificates, all in the frequency-support-shift regime. Increasing its cosine rank does not remove the general coverage deficit.

Table 4 Rates aggregated over three grid resolutions within each regime. The comparator is the best source-free design in that row.

Regime	Source feasible	Source certified	Best source-free certified	Difference
Aligned low frequency	100.0	56.7	Raw Sobol: 91.7	-35.0
Growing rank	100.0	40.0	Random low frequency: 20.0	+20.0
Frequency shift	95.0	60.0	Natural blockwise: 25.0	+35.0
Permutation	100.0	90.0	Random low frequency: 50.0	+40.0
Irregular grid	100.0	15.0	Raw Sobol: 38.3	-23.3
Piecewise smooth	100.0	95.0	Raw Sobol: 50.0	+45.0
Sparse high frequency	63.3	21.7	Natural blockwise: 25.0	-3.3
Misspecified target	75.0	0.0	Raw Sobol: 1.7	-1.7

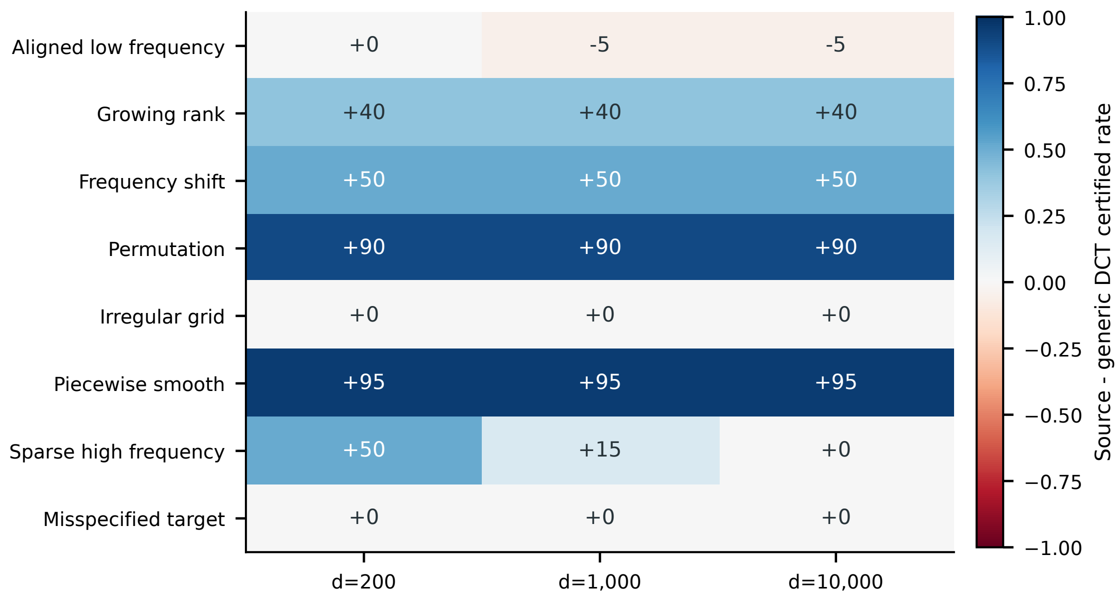

Figure 4 Source-minus-generic-DCT certified-success differences by regime and grid resolution. Positive aggregate performance coexists with clear reversals.

The aligned-low-frequency reversal has a concrete geometric explanation. Figure 5 reconstructs each fixed design and accesses the hidden target safe center only afterward. Across the 600 design points per arm, raw Sobol has median latent-center distance 0.0343 and median non-DC low-frequency energy 0.0258, compared with 0.0813 and 0.2292 for source scoring. Its pointwise true-feasible rate is 94.7%, versus 10.0% for both source and generic DCT. High-dimensional independent coordinates average toward a near-constant profile with DC coefficient .5; the aligned generator's safe center happens to occupy that region. This after-the-fact diagnostic explains the reversal, but it uses hidden geometry and cannot be used to choose a method or change an outcome.

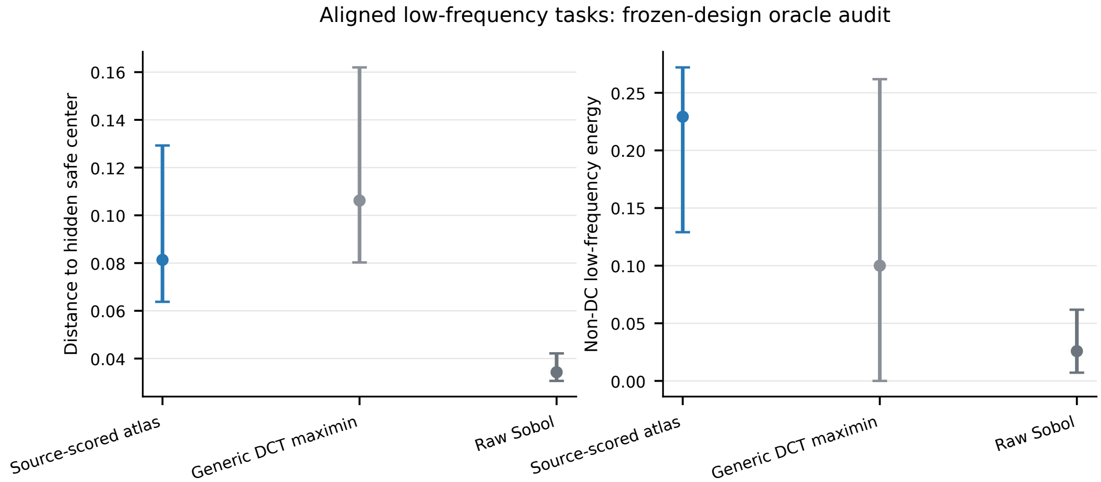

Figure 5 After-the-fact geometry diagnostic for the aligned regime. Points show medians and bars show 10th–90th percentiles over fixed design points. The hidden safe center was unavailable to every algorithm.

Table 5 Controls for the information available at $d = 1000$.

Schema	Source weighting	Tasks	Feasible (%)	Certified (%)	Loss
declared	domain blind	160	95.0	46.2	0.143
declared	conditioned	160	85.0	44.4	0.258
schema blind	domain blind	160	92.5	35.0	0.180
schema blind	conditioned	160	82.5	33.1	0.296

7.3. Schema and sensitivity

Declared profile ordering is valuable information. With domain-blind source weights, removing the declared schema reduces certified success from 46.2% to 35.0% (Table 5). Descriptor-conditioned weighting reduces both declared- and blind-schema performance and is not used in the final method. Source scoring therefore does not need the target family label, but it does rely on an interpretable ordered profile map.

Sensitivity is nonmonotone, which argues against treating source volume as a free advantage. Raising profiles per source task from 64 to 128 reduces certified success from 46.2% to 38.8%; using one, three, or five replications gives 45.0%, 46.2%, and 45.6%. Four source tasks give 45.0%, not more than two. Retaining 16 frequencies reaches 48.1%, while four reaches 43.1%. Increasing n_0 from 10 to 20 yields only 46.9% certification, while $n_0 = 5$ yields 42.5%. The first-center safety weight is more consequential: .25 reduces certification to 32.5%, whereas .5 and .75 give 46.2% and 45.0%. Figure 6 summarizes the source-budget axes.

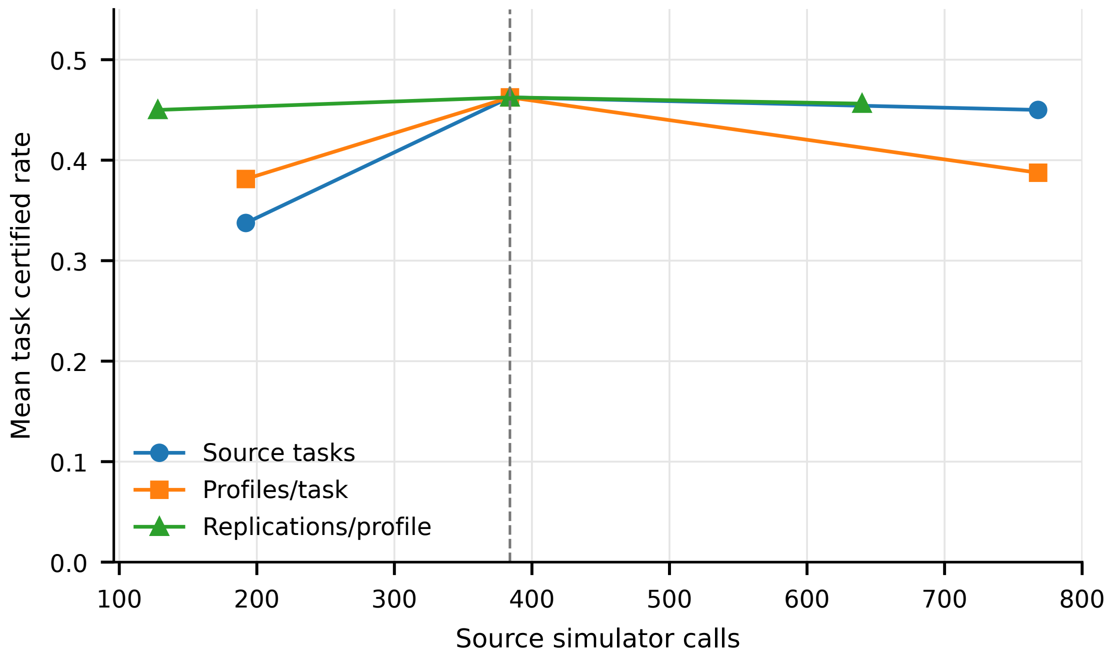

Figure 6 Source-budget sensitivity. More tasks, profiles, or replications do not monotonically improve the frozen design.

7.4. Cost changes the ranking

At the source method's primary budget, mean unamortized all-in calls are about 559 across the randomized law, while amortization over 20 targets reduces the source contribution from 384 to 19.2 calls per target. Comparable primary source-free designs use roughly 213–233 calls because verification dominates their ten search calls. Figure 7 makes both accounting views explicit.

When the 384 source calls are instead given to target-only search, the source method is not best (Table 6). It certifies 46.2% of 160 tasks. Generic DCT and raw Sobol each certify 53.8%, natural structure 53.1%, and target-only functional SCBO 58.1%. The latter includes one failed cell as an unsuccessful outcome. Mean realized all-in costs are already close because methods consume different verification effort. The source archive is thus an amortized multi-target investment, not a one-target total-cost win.

Using primary mean call counts, the source archive breaks even in simulator calls after roughly 7 target deployments versus generic DCT, 9 versus natural structure, and 11 versus raw Sobol. These are cost thresholds, not guarantees of outcome dominance; Energy demonstrates why the distinction matters.

Table 7 converts amortized calls into calls per true-feasible certificate. With the archive shared by 20 targets, source scoring uses 411.3 calls per certificate, versus 2192.9 for generic DCT, 1143.2 for

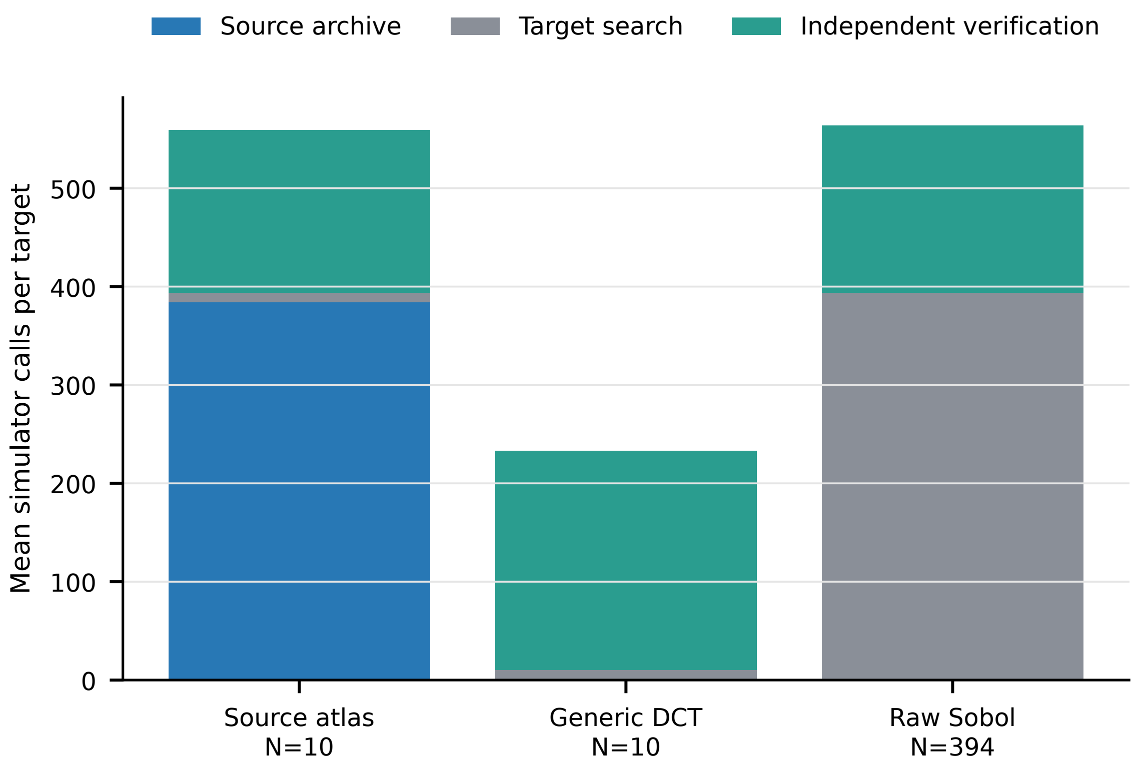

Figure 7 Mean source, target-search, and verification calls under unamortized and 20-target accounting.

Table 6 Equal preverification cost at $d = 1000$: 384 source plus 10 target calls versus 394 target-only calls. Verification remains additional and is included in the final column.

Method	Source	Target	Certified (%)	False	Algo. fail.	Mean all-in
Generic DCT maximin	0	394	53.8	0	0	564.5
Natural blockwise	0	394	53.1	0	0	564.5
Random low frequency	0	394	45.6	0	0	584.0
Raw Sobol	0	394	53.8	0	0	564.0
Source-scored atlas	384	10	46.2	0	0	562.0
Target-only functional SCBO	0	394	58.1	0	1	549.1

natural structure, and 709.4 for raw Sobol. This descriptive ratio does not price unsafe deployment, abstention, or objective quality.

7.5. External Energy result

Energy is a negative external control for source learning. Table 8 shows that source scoring certifies 60/90 runs, while generic DCT maximin certifies 70/90. Random low frequency, natural constants, raw Sobol, and functional SCBO certify 61, 57, 52, and 58. No arm records a false certificate.

Table 7 Outcome-adjusted simulator cost. Calls per success equal mean amortized all-in calls divided by the observed true-feasible certification rate. Break-even compares source scoring with each source-free row.

Matrix	Design	Certified (%)	Calls/success ($M = 1$)	Calls/success ($M = 20$)	Break-even M
$N = 10$ primary	Generic DCT maximin	10.6	2192.9	2192.9	1
$N = 10$ primary	Natural blockwise	19.4	1143.2	1143.2	2
$N = 10$ primary	Random low frequency	13.5	1716.9	1716.9	1
$N = 10$ primary	Raw Sobol	30.0	709.4	709.4	3
$N = 10$ primary	Source-scored atlas	47.3	1182.7	411.3	–
Equal preverification	Generic DCT maximin	53.8	1050.2	1050.2	2
Equal preverification	Natural blockwise	53.1	1062.6	1062.6	2
Equal preverification	Random low frequency	45.6	1280.0	1280.0	1
Equal preverification	Raw Sobol	53.8	1049.3	1049.3	2
Equal preverification	Source-scored atlas	46.2	1215.1	426.4	–

Table 8 Energy V3 region-held-out results over 18 markets and five seeds per market. Source scoring does not beat the generic structured control.

Design/backend	Certified	False	Median objective	Mean verify
Source-scored atlas	60/90	0	0.0345	145.8
Generic DCT maximin	70/90	0	0.0455	153.8
Random low frequency	61/90	0	0.0394	157.3
Natural constant grid	57/90	0	0.0391	174.2
Raw Sobol	52/90	0	0.0722	162.7
Target-only functional SCBO	58/90	0	0.0447	162.7

Across five regions, the source-minus-generic safe-rate difference is -0.079 with bootstrap 95% CI $[-.137, -.020]$: zero region wins, three losses, and two ties. Source scoring is not rescued by a favorable objective comparison; median objective is 0.0345 for source scoring and 0.0455 for generic DCT among certified summaries, but the sets of certified runs differ and conditional medians cannot override the coverage reversal.

The temporal audit preserves 312 of 358 V3 certificates under both block and nonoverlap conditions. It is descriptive, not an iid future-calendar certificate. The external conclusion is consequently narrow but useful: ordered low-frequency structure helps relative to raw Sobol, while source outcomes can mis-rank that structure under geographic and temporal shift.

7.6. Verifier power and native transfer

The verifier is valid but intentionally conservative. At 80 all-success replications, certification power is only 1.7% at the threshold $p = .95$, 13.2% at .975, and 44.8% at .99 (Figure 8). Therefore the gap between 91.7% feasible coverage and 47.3% certified deployment is expected; an empty certificate

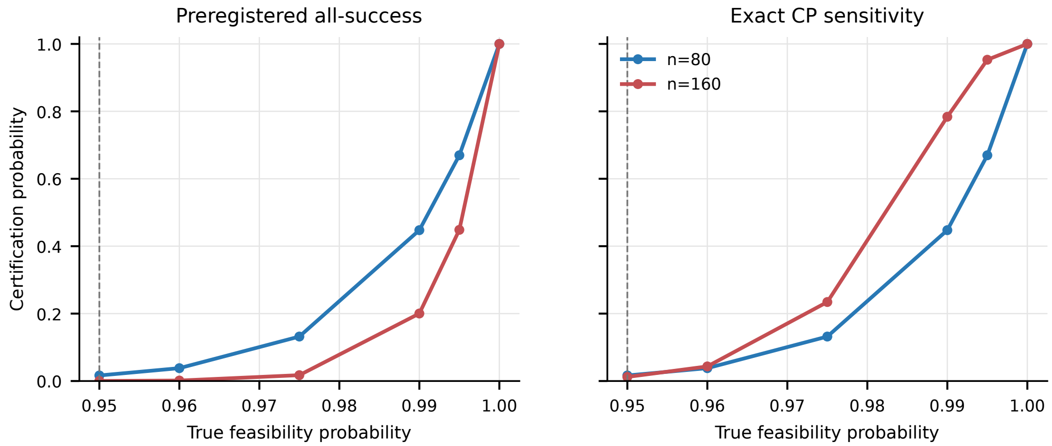

Figure 8 Exact fixed-budget certification power. The preregistered all-success rule is familywise valid but weak near the chance threshold.

set is not evidence that search found no feasible point. A Clopper–Pearson threshold rule is reported as sensitivity, not substituted after observing outcomes.

In the native transfer matrix, RGPE certifies 30/60 tasks and MetaBO 28/60; the other six methods range from 0/60 to 14/60 (Online Supplement, Table EC.4). All eight certify 0/20 Queue tasks. These results show that ordinary end-to-end transfer remains difficult, but they do not establish a broad SOTA ranking from three historical synthetic families.

Finally, cumulative factor-HVD substantially improves held-out log-variance RMSE and variance-shape correlation relative to pooled variance, yet does not improve feasible recovery, regret, false certification, or verification cost under matched designs. It is therefore retained only as a calibration diagnostic in the Online Supplement, Section EC.3, not as a main optimization contribution.

8. Discussion

8.1. What the evidence supports

Source outcomes help when source and target tasks organize safety around similar policy shapes. That is the central empirical claim, and the adverse regimes make it testable rather than automatic. Source scoring improves coverage on the full randomized mixture, but the regime table and Energy study show concrete cases in which it loses.

This interpretation differs from claiming a generic high-dimensional optimizer. The raw vector may contain 10^4 coordinates, but the decision is an ordered profile and the library has finite structural complexity. A controller may still require a fine lookup table, so profile-grid refinement is operationally relevant. It is not equivalent to adding 10^4 unrelated decision variables. The proper benefit is invariance to representation resolution under shared profile structure.

8.2. What Energy teaches us

The Energy reversal shows that a generic structural design can beat source scoring when rankings from held-out regions do not transfer. Because V3 was fixed before its outcomes, changing the score now would turn a test case into development data. We keep the result as a negative control and as a practical boundary on when the method should be used.

The practical implication is a diagnostic choice. If historical tasks and a new target share an ordered policy semantics and source rank stability can be validated on genuinely held-out tasks, source scoring is reasonable. If the target is temporally or geographically shifted and generic low-frequency structure already has a credible physical basis, the generic maximin design is the safer default. The method should abstain from a transfer claim when these conditions cannot be assessed.

The current method begins after this assessment: a source archive has already been supplied, and the algorithm ranks profiles inside it. It does not retrieve related tasks from an unrestricted historical database. Regime-matched synthetic sources therefore identify the value of scoring a relevant archive, whereas Energy shows the cost of archive mismatch. Source retrieval and its own data-acquisition cost remain separate research and operational decisions.

8.3. When the source archive pays for itself

Ten target calls do not beat 394 target calls for a single target. The source archive becomes economically plausible only when it is reused. Let $C_0 = N_0 + V_0$ be the expected target-only calls per deployment and $C_A = N_A + V_A$ the source-design calls excluding the archive. The call-count break-even number of deployments is

$$M_{\text{break}} = \left\lceil \frac{S}{C_0 - C_A} \right\rceil, \qquad C_0 > C_A. \tag{23}$$

In the randomized primary experiment, mean-call estimates give thresholds of about 7, 9, and 11 targets relative to generic DCT, natural structure, and raw Sobol. This arithmetic does not include differences in simulator setup, wall-clock parallelism, or policy quality, and Energy shows that paying the archive can still be inferior. It is a planning threshold, not a dominance theorem.

Table 7 also reports calls per certified deployment, but that ratio is only a descriptive planning aid. A complete operational comparison would attach decision-maker costs to unsafe deployment, abstention, and objective quality. We report those components separately rather than choose their relative value on the reader's behalf.

8.4. Search and verification answer different questions

Independent verification frequently costs an order of magnitude more than target search because it answers a different question. Search asks whether a promising policy has been found. Verification asks whether a fixed policy can be deployed with a specified false-certification bound. A target-only optimizer making the same safety claim must also pay for fresh samples.

The all-success rule is valid but weak near the threshold, which explains why feasible coverage exceeds certified deployment. Future studies should fix a more efficient exact or sequential test in advance and use block-aware verification when temporal replications are not iid.

8.5. Benchmark fit and remaining threats

Fixing the generator before confirmatory runs reduces development overfitting but cannot erase it: the task law, library, and stress axes were created within the project. Ordered semantics are a strong prior. Energy covers only five regions, and the native transfer study has only three historical families. The archive is assumed to have been supplied; source retrieval is unsolved. Three replications cannot order near ties, source aggregation can hide task disagreement, and a finite library can miss the safe region entirely. Regret uses a finite evaluation library rather than the unknown continuous optimum, and one functional-SCBO failure remains in the denominator. The theory is conditional on alignment, independence within each regime, and the declared replication law. Finally, the same target seeds appear at all three grid resolutions, so 480-cell totals are descriptive rather than 480 independent tasks.

9. Conclusion

A policy stored as thousands of ordered numbers may still be one simple rule. We use source simulations to rank transparent policy shapes, farthest-first selection to keep ten starting policies diverse, and fresh simulations to make the final safety decision. This improves coverage on the declared task population but loses under adverse transfer, in Energy, and at equal single-target cost. Historical simulation is therefore a reusable investment for related ordered-profile problems, not a shortcut for an arbitrary high-dimensional black box.

10. Code and Data Disclosure

Code and the declared seeds generate every synthetic instance. The Energy study uses the public data cited in Section 6. An anonymous package provides code, fixed manifests, compact results, and build instructions; raw checkpoints are unnecessary. A persistent identifier will replace the anonymous location upon acceptance.

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